

Aho-Corasick Backend API Documentation with Results

1. Add Keywords API

Endpoint:

POST /keywords

Functionality:

- Add multiple keywords/patterns to the system.
- Keywords are inserted into the Trie structure.
- Failure links are generated for efficient searching.

Example Request Body:

```
{  
  "keywords": ["spam", "scam", "fraud"]  
}
```

The screenshot shows an API client interface for a POST request to the endpoint `/keywords`. The header bar indicates the method `POST` and the endpoint `/keywords` with the label `Add Keywords`. Below the header, there is a `Parameters` section with a `Cancel` button and a `Reset` button. The `Parameters` section shows `No parameters`. The `Request body` section is marked as `required` and has a dropdown menu set to `application/json`. Below this, there is a text area for the request body with the JSON content: `{ "keywords": ["spam", "scam", "fraud"] }`. At the bottom, there is a blue `Execute` button and a `Clear` button.

Figure 1: POST /keywords API Request



Figure 2: POST /keywords API Response

2. Search API

Endpoint:

POST /search

Functionality:

- Search input text using the Aho-Corasick algorithm.
- Return matched keywords.
- Return positions and counts of matches.

Example Request Body:

```
{
  "text": "This message contains spam and fraud."
}
```

POST /search Search

Parameters

No parameters

Request body required application/json

Editor Value | Schema

```
{
  "text": "This message contains spam and fraud."
}
```

Execute Clear

Figure 3: POST /search API Request

Responses

Call

```
curl -X POST \
  http://localhost:8080/search \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "text": "This message contains spam and fraud."
  }'
```

Request URL

http://localhost:8080/search

Server response

Code	Details
200	Response body <pre>{ "matches": [{ "keyword": "spam", "position": 0, "count": 1, "length": 4 }, { "keyword": "fraud", "position": 0, "count": 1, "length": 4 }], "total_matches": 2, "unique_keywords_found": ["spam", "fraud"], "text_length": 33, "keywords_searched": 1 }</pre> Response headers <pre>access-control-allow-origin: http://localhost:8080 content-length: 218 content-type: application/json date: Fri, 22 May 2020 11:29:35 GMT server: node vary: Origin</pre>

Figure 4: POST /search API Response

Example Response:

```
{
  "matches": [
    {
      "keyword": "spam",
      "position": 22
    },
    {
      "keyword": "fraud",
      "position": 31
    }
  ]
}
```

3. Health Check API

Endpoint:

GET /health

Functionality:

- Verify whether the API server is running correctly.
- Show the number of keywords currently loaded in memory.

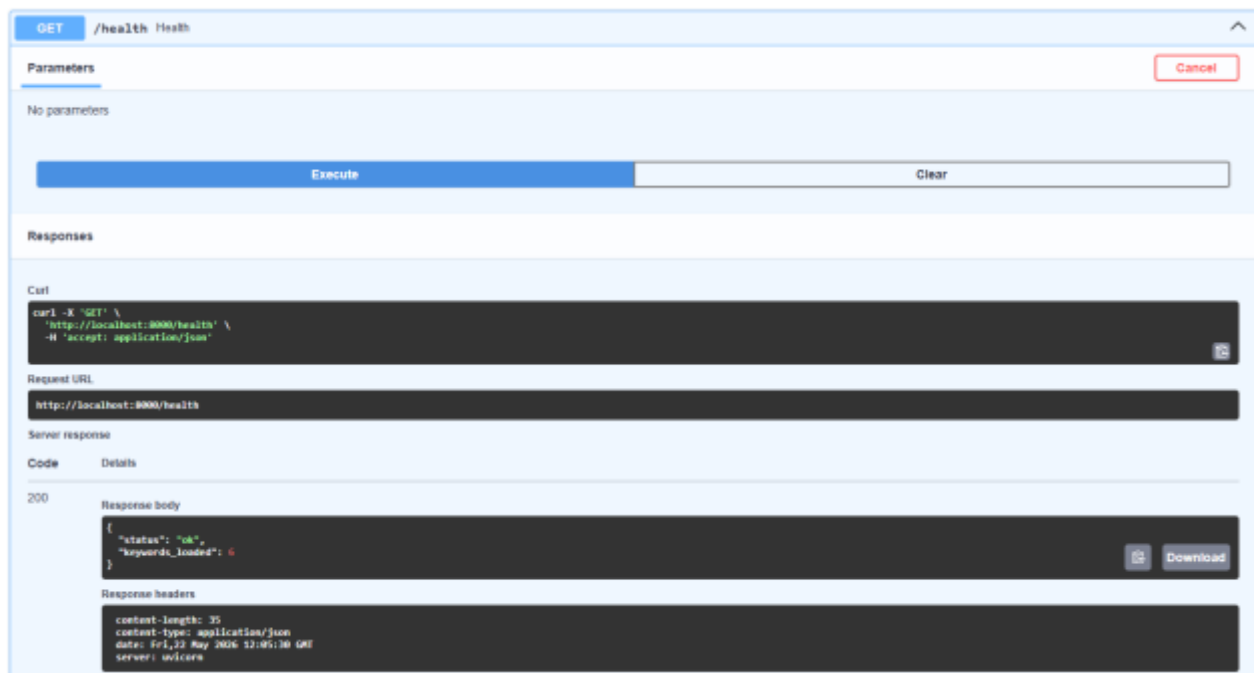


Figure 5: GET /health API Response